



6 steps to success with generative AI

Public sector's leader's guide to driving
tangible value quickly



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Embracing the generative AI opportunity

Generative AI is rapidly moving from the latest breakthrough technology to an everyday reality. Organizations of all sizes across all industries are using gen AI to unlock growth opportunities, boost employee productivity, and accelerate process optimization. In fact, Gartner predicts that roughly 80 percent of enterprises will have deployed gen AI applications by 2026.¹

As a powerful change agent, gen AI raises key questions for leaders about where it fits in their organization and how to use its full potential. The lack of a single playbook for gen AI success is keeping some leaders on the sidelines, unsure of how to get started or where to go next.

Amazon Web Services (AWS) is helping tens of thousands of customers build and scale with gen AI to deliver value, and we've identified common steps that companies of all sizes can use to move quickly and confidently with secure gen AI. This ebook outlines that proven path—from taking the first step to measuring results—with insights from real-world examples and AWS experts to help you drive meaningful impact with gen AI.



¹ "Gartner Says More Than 80% of Enterprises Will Have Used Generative AI APIs or Deployed Generative AI-Enabled Applications by 2026," Gartner, October 2023

² "The State of Generative AI," AWS Executive Insights, 2023

GEN AI ECONOMIC IMPACT POTENTIAL

\$2.6T–\$4.4T

Gen AI's impact on productivity could add \$2.6 trillion–\$4.4 trillion in value to the global economy²

What are artificial intelligence, machine learning, and generative AI?

AI is a way to describe any system that can replicate tasks that previously required human intelligence. Almost all AI systems are built using machine learning (ML). ML uses large amounts of data to create and validate decision logic, forming the basis of an AI model. The AI application then feeds input data into that model, and the model outputs humanlike decisions. The rapid advancement of ML, massive proliferation of data, and easy availability of scalable compute capacity can help organizations accelerate how they use AI and, particularly, how they use gen AI.

Gen AI is a type of AI that can create new content and ideas, including conversations, stories, images, videos, and music. Like most AI, gen AI is powered by ML models—very large models that are pre-trained on vast amounts of data and commonly referred to as foundation models (FMs).



Why generative AI?

Gen AI has quickly emerged as a game-changing technology with the potential to create significant impact across a wide range of applications. From enhancing user experiences to boosting employee productivity to accelerating process optimizations, organizations are using gen AI to gain an edge and unlock new sources of value.

For convenience, we can group the potential impact of gen AI into three types:

- **Enhances user experiences** by providing a conversational interface to access information and applications by employees and end users
- **Boosts employee productivity** by helping them create content like conversations, images, codes, videos, and more
- **Optimizes processes** by extracting insights from large amounts of documents to improve decision-making and knowledge sharing

Generative AI use cases across industries

Enhances user experiences

- Chatbots and virtual assistants
- Agent assist and conversational analytics
- Hyper-personalization

Boosts employee productivity

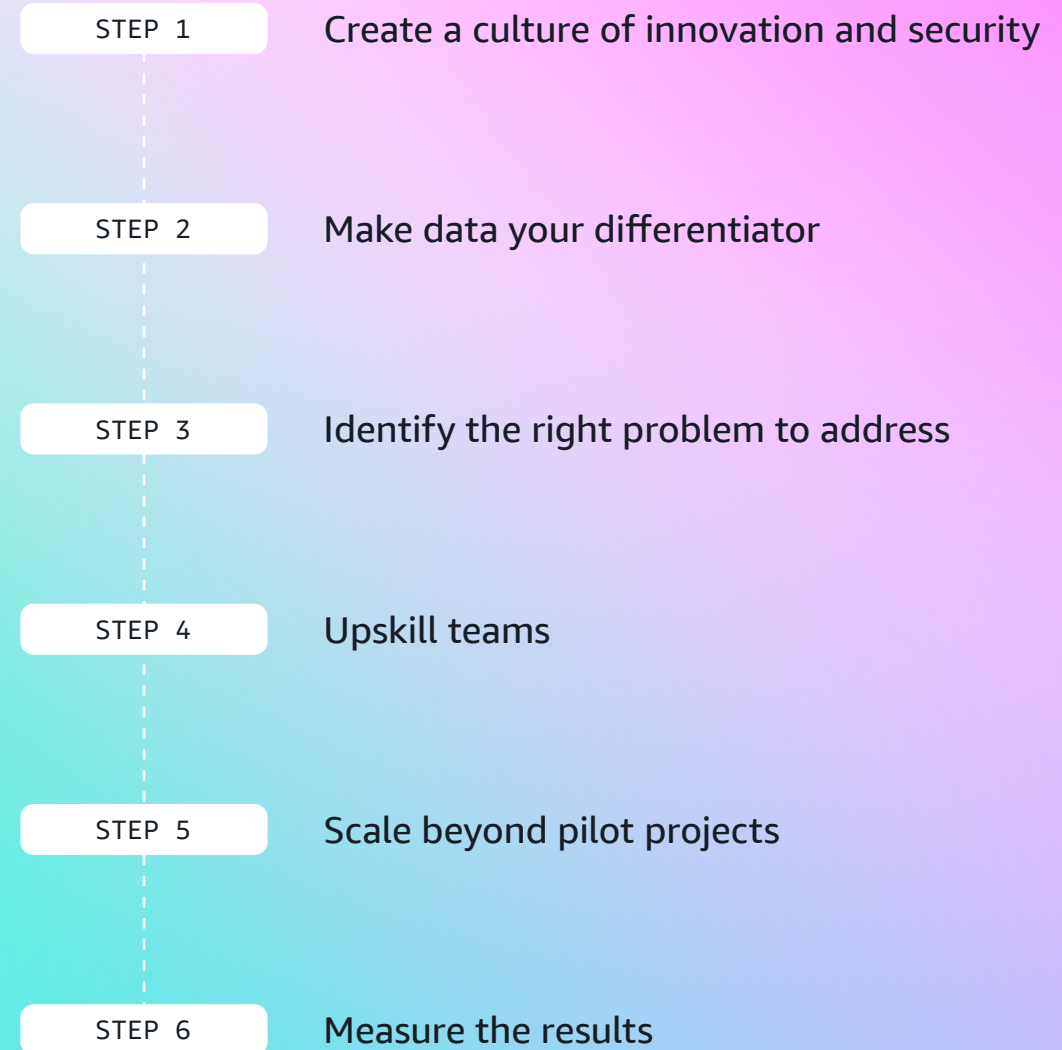
- Coding assistant
- Content discovery
- Content creation
- Conversational business intelligence (BI)
- Text summarization

Optimizes processes

- Intelligent document processing (IDP)
- Fraud detection
- Supply chain management
- Identity verification

The generative AI journey

The gen AI journey is different for every organization and not necessarily a straightforward path. Achieving success with gen AI requires more than great technology—it also means making sure the organization is aligned to the right goals. Identifying and reaching those goals will necessitate broad changes in processes, management, and culture. The next sections will explore how organizations can overcome common challenges that impede progress and take the right steps to implement AI in efficient, sustainable ways.



STEP 1

Create a culture of innovation and security

Unlocking the full potential of gen AI requires cultural changes in team organization, objectives, and outlook.

For gen AI to proliferate through your organization, both operations and technical teams must work together and share the same priorities. To achieve this at the outset, your gen AI effort must be supported from the highest levels, with goals set by executive champions and an investment in the technology and processes that enable success. This includes championing a commitment to building responsibly from the beginning—working to identify and mitigate bias, improve explainability, and help keep data private and secure. Your successful gen AI effort further requires including layers of human review and building guardrails to help adhere to security and responsible AI principles. By taking a people-centric approach, your organization can work to educate its workforce on responsible AI and build diverse teams to bring more perspectives to the table and improve fairness.

Making security an enabler of generative AI innovation

Security and privacy must be built in from day one. Investment in a strong security culture is what enables organizations to embrace emerging technologies safely and responsibly.

For example, highly regulated sectors like financial services, healthcare, and life sciences have proven most adept at capitalizing on gen AI swiftly and safely. Their sustained investments in compliance rigor and a pervasive security culture positions them to precisely assess risk appetite, adhere to control frameworks, and confidently harness innovations at the speed their organizations demand.

Embracing an iterative, experimental mindset

It's crucial for management to have a broad perspective while fostering AI initiatives. Executives must establish clear goals but remain flexible in the organization's approach to achieving them. Like any journey, there will be missteps and detours along the way. Companies that embrace these learning opportunities and apply the insights gained can foster an innovative culture throughout the business. Constant experimentation is essential for success with gen AI.

Upskilling the workforce for the generative AI era

As new gen AI applications are deployed, organizations should provide training to employees on how to use gen AI responsibly and effectively. Organizations should also have clear policies guiding appropriate usage, while leaders should encourage experimentation with gen AI while managing risks.

How Amazon transformed its business with generative AI

Amazon has been at the forefront of AI innovation for decades. As an early pioneer, Amazon used AI to reinvent shopping experiences, provide entertainment, optimize operations, and drive innovation across our businesses.

Amazon uses gen AI powered by AWS to deliver new value to customers. From our earliest days, Amazon has used AI to change the way customers shop. With over [8,600 products sold every minute](#), AI-driven innovations like [personalized recommendations](#) help customers effortlessly discover products tailored to their unique interests and preferences.

Amazon also uses gen AI to build a sustainable business for customers and the planet. By optimizing the computing power, data storage, and networking needed to run gen AI applications, Amazon reduces its environmental impact. In addition, the company is creating tools and guidelines to help its customers make their gen AI applications more sustainable.

From smarter assistants to automated ad creative to streamlined healthcare access and more, Amazon uses gen AI to drive innovation.



Here are a few examples:



Amazon created [Amazon Rufus](#), a gen AI-powered conversational shopping assistant that makes it easier to discover, research, and buy products in the Amazon shopping app, ultimately helping customers make more informed purchase decisions.

ads

[Amazon Ads](#) uses gen AI to help advertisers and brands build successful ad campaigns and elevate the customer discovery experience by creating engaging and vibrant images.

pharmacy

[Amazon Pharmacy](#) uses [Amazon Bedrock](#) and [Amazon SageMaker](#) to provide upfront pricing estimates for 99 percent of prescriptions and optimize operational efficiency, helping customers access needed medications faster.

amazon

Through [Project P.I.](#), Amazon uses gen AI and computer vision to spot damaged products before they're shipped to customers. Also, Amazon integrated its own gen AI assistant, [Amazon Q Developer](#), to transform software development, which has helped them save \$260M and 4.5K developer-years of work.³

amazon music

And Amazon launched [Maestro](#), a new feature on Amazon Music that uses gen AI to make it easier for listeners to build playlists of songs they're likely to enjoy.

[Read all the stories](#) →

\$260M

With Amazon Q Developer,
Amazon has saved \$260M and
4.5K developer-years of work³

Make data your differentiator

As with ML, data is an important piece of gen AI. While general large language models (LLMs) may be used “out of the box” for some use cases, when you want to build gen AI applications that are unique to your business needs, your organization’s data is your strategic asset. Your data is the difference between a generic gen AI application and one that truly knows your business and your customer to deliver a better, more differentiated experience.

While some organizations will build and train their own LLMs with vast amounts of data, many more will use their organizational data to fine-tune FMs for their unique business needs or to add context to prompts through Retrieval Augmented Generation (RAG). For example, you can use data from sources such as your data lake, database, and data warehouse to create a chatbot that can provide technical support for your offerings or a model that provides email copy trained on your best-performing campaigns.

Because of this, success with gen AI requires relevant, high-quality data, which means that you need a strong data strategy in the cloud. According to [McKinsey Digital](#), “...companies that have not yet found ways to effectively harmonize and provide ready access to their data will be unable to fine-tune gen AI to unlock more of its potentially transformative uses.”

The right data strategy for gen AI includes a comprehensive set of services to store and query data at scale, breaks down silos so you have ready access to all your data to leverage for gen AI applications, and makes sure your data is secured and governed throughout the lifecycle of building gen AI applications.

AWS makes it easy to use your data to customize and augment FMs to suit them to your specific needs and offer unique experiences to your users. AWS offers a comprehensive range of data services to store, process, manage, and secure your data, enabling you to power your gen AI applications using AI-ready data.

Discover how a [data foundation built on AWS](#) gives you a strategic advantage when it comes to gen AI.

“We’re thrilled to partner with AWS to make generative AI more accessible and sustainable for our agencies. This allows us to harness generative AI’s potential while responsibly managing resources.”

[Jeffrey Chai](#), Product Manager of MAESTRO, Singapore Government Technology Agency.

STEP 3

Identify the right problem to address

Before you can successfully adopt gen AI, it's important to determine where and how you can apply the technology for the greatest impact. In other words, you need to select the right gen AI use case for your organization. This requires the consideration of several factors.

First, look for a problem that needs solving. Ask yourself: What challenge is most critical to your success? Pick the areas of your organization that need attention. In general, your initial use case or cases should be relevant to your organization's success, viable, and impactful.

Next, choose a pilot that is low risk, can easily be implemented at the right cost, and delivers a fast, measurable return on investment (ROI). These are often use cases focused on internal processes or workforce productivity. By showing value and ROI quickly, you can build and maintain momentum for further use of gen AI throughout your organization.

In helping tens of thousands of customers drive value with gen AI, we've identified the top three gen AI use cases that are helping organizations drive impact and momentum. Download the [3 Leading Generative AI Use Cases for the Public Sector](#) ebook for more guidance on gen AI use cases.

Once you've aligned on a use case, it's important to include both technical and domain experts. Technical experts need the field knowledge of domain experts to define precise challenges and identify the data most important to finding a solution. This approach is also critical to change management. When technical and domain experts collaborate to create models, employees will feel more confident in making decisions based on the output. Together, these teams should also work through how to measure success.

For more on measuring the success of AI initiatives, refer to [Step 6 in this eBook](#).





GovTech boosts public sector productivity with Amazon generative AI

The [Government Technology Agency](#) (GovTech) of Singapore is a statutory board responsible for delivering digital services to the public. As the Centre of Excellence for Infocomm Technology and Smart Systems, GovTech also develops the Singapore government's capabilities in data science, including generative AI.

Opportunity

GovTech identified generative AI as a way to drive innovation, enhance public services, and improve lives within the community. However, the high costs of operating LLMs at a government scale were key considerations for the organization.

Solution

To address cost concerns and tap into the potential of generative AI, GovTech collaborated with the AWS Generative AI Innovation Center to develop generative AI solutions that enable cost-effective adoption of this transformative technology across government agencies. By using [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing FMs from leading AI companies, and Amazon SageMaker JumpStart, an ML hub with FMs, built-in algorithms, and pre-built ML solutions, GovTech improved cost performance for generative AI workloads by up to 75 percent.

The Central Provident Fund Board (CPF Board) of Singapore, a statutory board whose mission is to enable Singaporeans to have a secure retirement through lifelong income, healthcare financing, and home financing, used opensource, quantized generative AI models deployed through MAESTRO to summarize call transcripts for the ~600,000 calls that it receives on average each year. These summaries are vital for follow-ups, investigations, and quickly identifying emerging issues.

[Read the full story](#) →

Upskill teams

Today's organizations are increasingly focused on arming their teams with the right skills to succeed in the era of gen AI. That includes not only technical roles but also development, marketing and communications, and even the CEO. A 2024 KPMG survey found that over two-thirds of organizations are currently training their workforce in gen AI, and 61 percent are actively hiring new talent to support their gen AI initiatives.⁴

To help individuals increase innovation, efficiency, and creativity in their current roles—and train for the AI jobs of the future—AWS offers the [Generative AI Learning Plan for Developers](#)—a series of free on-demand training sessions designed to introduce gen AI to software developers. Participants will learn how to plan a gen AI project and get started with [Amazon Bedrock](#) while exploring the foundations of prompt engineering and the architecture patterns to build gen AI applications. Additionally, AWS offers training for a variety of roles through more than 600 free, on-demand cloud courses in up to 14 languages on [AWS Skill Builder](#).

While there's no one-size-fits-all solution to the AI skills gap, there are proven methods that can maximize the abilities of existing staff, reducing the need to make large investments in buying or borrowing pretrained expert talent.

These methods include:

- **Defining the skills gap:** Before closing the skills gap, an organization must identify the precise differences between what it needs or wants its staff to do and what its employees currently can do. AWS offers a free [AWS Learning Needs Analysis](#) to identify skills gaps and design an effective learning plan.
- **Understanding how skills are mapped:** Because AI initiatives are interdisciplinary efforts, an organization should map the skills needed across data scientists, data engineers, analysts, application developers, statisticians, and other subject matter experts in the business.
- **Customizing training for specific needs:** Most organizations have training programs. Leaders need to identify the skills needed for their organization's specific AI requirements and add the appropriate training to their programs.

31M

AWS has helped more than 31 million learners across 200 countries and territories build their cloud skills through free training initiatives



Organizations also need to align teams to successfully tackle generative AI problems by:

1. Promoting a culture of empowered teams

AI project teams must be cross-functional, with the authority to execute individual objectives and the freedom to organically cross-pollinate with other teams as demands dictate and opportunities arise. To make this kind of teamwork possible, management will need to embrace new structures—letting go of the strictly hierarchical and departmentally siloed organizational models of the past.

2. Starting with a pilot team

They should establish a pilot team of engineers, IT and AI practitioners, and program leaders and task it with an AI project. It's also critical to have an internal champion who promotes a culture of continuous learning. Having top-down buy-in is key to moving your gen AI project from proof of concept to production.

3. Enabling organic transformation

Once the pilot project is complete, the organization can split up the team, add people to create new teams, and task them with new projects. This process continues, allowing knowledge to spread organically from veteran team members to new recruits and pollinate between teams.

By following this guidance, many organizations are finding that the people they currently have are the people they need to close their AI skills gap. While some recruiting may still be required, investment in the right tools, processes, and management changes can do much of the work to upskill talent for AI success.



How Terrascope boosts decarbonization efforts by upskilling with AWS

[Terrascope](#) is a global, end-to-end decarbonization solution that supports organizations in their green energy transformation through intelligent technology, deep decarbonization experience, and climate science expertise.

Opportunity

Terrascope opted to use gen AI to help its customers solve the data management, emissions measurement, and decarbonization planning challenges associated with their sustainability initiatives. Seeking to derive maximum value from the technology, Terrascope entered into an agreement with AWS to upskill more than half of its team members through [AWS Training and Certification](#) programs.

Solution

Training on the intricacies of AI technology and ML offerings, such as [Amazon SageMaker](#), has proven valuable for Terrascope. Since teaming with AWS, Terrascope has reduced deployment time by 97 percent and cloud spending by 30 percent. The company also enabled one of its customers to measure emissions five times faster than traditional processes.

“Our collaboration with AWS experts is helping us democratize knowledge of AI and cloud capabilities among our teams, enabling us to refine our ML models and deliver innovative solutions to customers.”

Mathavan Arugalaimuthu, CTO, Terrascope

[Read the full story](#) →

Scale beyond pilot projects

As our customers look to take successful proofs of concept and move them into production, they face numerous challenges that can hinder their ability to scale gen AI. In fact, Gartner estimates that at least 30 percent of gen AI projects will be abandoned after proof of concept by the end of 2025 due to poor data quality, inadequate risk controls, escalating costs, and unclear business value.⁵

Companies face several key challenges including:

1. **Identifying the right FMs:** With so many different FMs to choose from, companies struggle with finding the ones that fit their specific use cases and data
2. **Balancing performance and cost:** Organizations need to find cost-effective FMs that still deliver the required level of performance for their business needs
3. **Keeping pace with rapid advancements:** The fast-moving gen AI landscape means organizations require flexible solutions that can easily integrate the latest model improvements and capabilities
4. **Prioritizing security and responsible AI:** As organizations scale gen AI, they must address critical concerns around data security, responsible AI, and maintaining the accuracy and trustworthiness of the models

[Amazon Bedrock](#) has empowered many organizations to quickly and securely scale gen AI adoption. Using the latest high-performing FMs, Amazon Bedrock integrates cutting-edge AI advancements through a single API. Its robust orchestration tools streamline complex workflows and enhance operational efficiency. And its serverless architecture provides secure, high-quality, low-latency, and cost-effective deployments, integrating seamlessly with other AWS services.



With Amazon Bedrock,
you and your technical team can:

Innovate faster with a broad choice of models

Rapidly evaluate and select the optimal models for your unique use cases, making sure that your gen AI applications deliver maximum impact.

Gain a competitive advantage

Amazon Bedrock provides powerful tools so you can build highly differentiated gen AI applications to deliver personalized customer experiences using your own proprietary data, automate complex workflows to drive efficiency, and scale with the best price performance.

Build responsible gen AI applications

Safeguard your sensitive data and maintain the highest levels of trust and compliance with the robust security and governance features of Amazon Bedrock.



Sírio-Libanês hospital boosts reporting efficiency with Amazon Bedrock

[Sírio-Libanês Hospital](#), a healthcare institution in Brazil, is using Amazon Bedrock to revolutionize its reporting processes. With a focus on innovation and excellence, the hospital is harnessing the power of generative AI to streamline its operations and improve decision-making.

Opportunity

The hospital's Cloud Center of Excellence (CCoE) was facing a significant challenge in providing timely and accurate reports to various stakeholders. The manual process of gathering and analyzing data was time-consuming, taking up to several hours to complete. The CCoE team recognized the need for a more efficient solution that could automate reporting and free up resources for more strategic initiatives.

Solution

The hospital's CCoE team collaborated with AWS to develop Cloudinho, a generative AI agent based on Amazon Bedrock. Cloudinho is integrated with Microsoft Teams, the hospital's official communication tool, and uses AWS Cloud services to generate reports in minutes. The solution has reduced reporting time by 99 percent, from 3 hours to just 1.5 minutes.

"The project was possible because AWS gives us the best structure to work with digital assets. Working together is key here. The ideas are incubated here, but the guidelines with the best paths come from the AWS team and are fundamental to build and evolve these products."

Vitor Bellot, Coordinator of Operational Excellence at the Sírio-Libanês Hospital

[Read the full story](#) →

STEP 6

Measure the results

Measuring the success of gen AI initiatives requires a tailored approach focused on defining relevant key performance indicators (KPIs) that align with specific organization objectives.

By working closely with stakeholders to identify the right KPIs upfront can demonstrate meaningful, tangible value from your gen AI projects.

These KPIs might include metrics like:

- Increased staff productivity through the use of AI-generated content and analysis
- Improved user satisfaction and engagement from personalized AI-powered experiences
- Enhanced operational efficiency as a result of automating complex tasks and workflows
- Accelerated time to launch for new offerings and services leveraging gen AI

The key is to move beyond generic, financial-focused ROI and, instead, define a balanced scorecard of KPIs that reflect the unique impact of gen AI for your organization. This allows you to track and communicate the very real value being delivered, which can then justify further investment and expansion of your gen AI capabilities.





How Bayer Crop Science boosts employee productivity with Amazon Q

[Bayer Crop Science](#), a division of the multinational Bayer AG, sees gen AI as a key catalyst for next-level productivity. The company is enabling thousands of its data scientists and engineers to innovate agricultural solutions for farmers across the globe.

Opportunity

From planning to harvest, Bayer Crop Science strives to offer products and information that help farm operations reach their maximum potential. As modern agriculture becomes increasingly complex, Bayer is focused on leveraging gen AI to deliver gen AI-powered capabilities for its customers and making it easier for data scientists to develop innovative solutions faster that improve farm performance.

Solution

Enabled by gen AI on AWS, the Bayer Crop Science division is boosting employee productivity as it transforms software development. By leveraging [Amazon SageMaker Studio](#), its teams can rapidly build, deploy, and train ML models for various use cases to bring data solutions to market faster. They are also using [Amazon Bedrock](#) to power gen AI capabilities that reduce the support burden of complex, highly orchestrated workloads. When something fails, Amazon Bedrock aids in diagnosing issues with less engineering support than previously required. The Bayer division recognizes the importance of empowering its platform builders to focus on tasks that provide differentiated value. By implementing [Amazon Q Business](#), its developers can generate documentation and access operating information more efficiently, reducing onboarding time by up to 70 percent. And with [Amazon Q Developer](#), they can automate coding tasks, helping the company improve developer productivity by up to 30 percent.

“The technologies that are being created by AWS help accelerate and improve our company’s ability to produce food and fuel for a growing population. Our focus is moving rapidly toward leveraging gen AI to take us into the future.”

Will McQueen, VP, Head of Crop Science Data Assets & Analytics, Bayer Crop Science

[Read the full story](#) →

Solving the biggest generative AI challenges

Most organizations have explored the possibilities of gen AI. But many find themselves hitting barriers along the way, worried that costs and complexities will grow too high as they progress.

Throughout this ebook, we explored the steps to forge ahead and realize the full power of gen AI. To recap, let's look at the biggest challenges we identified along the way, along with a brief recommendation of how your organization can solve them.

To learn more about how you can overcome obstacles and accelerate your gen AI journey, visit the [AWS Generative AI Resource Hub](#).

[Learn more →](#)

Challenge	Solution
Discouragement from failures	Developing a fault-tolerant culture
Data quality	Creating a modern data strategy that includes data lakes
Finding the right problems	Building blended teams that include both technical and domain experts
The gen AI skills gap	Adopting new organizational models, processes, and team management philosophies
Sustainably scaling beyond pilot projects	Using fully managed services like Amazon Bedrock to gain access to high-performing models to quickly build and scale gen AI applications
Measuring the results	Forgoing traditional ROI in favor of agility and competitive advantage

Taking the next step with AWS

Gen AI is reshaping the future by giving organizations the opportunity to innovate continuously, deliver exceptional user experiences, and drive sustained growth.

Join thousands of AWS customers using our secure, flexible, and scalable AI services to redefine how work gets done. AWS delivers the most comprehensive set of capabilities at every layer of the stack to help you innovate with gen AI, including the most performant and cost-effective infrastructure for training and running ML models at scale, the easiest way to build gen AI applications with LLMs and other FMs, and the most capable gen AI-powered assistant to help transform how work gets done. Plus, all of our gen AI offerings come with enterprise-grade security and privacy built in, delivering confidence and peace of mind.

AWS Generative AI Innovation Center

The [AWS Generative AI Innovation Center](#) pairs customers with AWS science and strategy experts with comprehensive expertise spanning the entire gen AI journey. This program helps customers build a strategic and holistic approach aligned with business goals, use cases, and operational challenges.

Get started now →

